BACHELOR'S THESIS (TFG) · UC3M

Four papers for RL in CARLA

Bullet-point summary

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Numbers come from the PDFs in `papers/`. Items marked *(my proposal)* are ideas for CARLA, not results from the papers.

Why these four papers

- CARLA detects cardilla in DJI drone video; the previous thesis trained YOLO-nano once (mAP50 0.81–0.85).
- This TFG: after every uploaded flight, a person labels a few tiles and the detector is updated.
- The loop needs four answers, one per paper:
  - **Which tiles to label?** → TAILOR (the RL component).
  - **How to use the labels + unlabeled footage?** → Simple Supervision (RTSM).
  - **How to retrain every round?** → On Warm-Starting (Shrink & Perturb).
  - **How to compare sunny and cloudy flights?** → LUF-net (colour correction).
- Only TAILOR is RL (a multi-armed bandit). The other three make the loop work.

1. TAILOR — NeurIPS 2023

Problem

- Many active-learning strategies exist; which one is best changes per dataset and cannot be known in advance.
- Earlier online selectors used training-accuracy rewards, which are useless for deep networks (always near 100%).

Method

- Each labeling strategy is an arm of a multi-armed bandit.
- For each slot in a batch, Thompson sampling (Beta posteriors) picks an arm; that strategy picks one example.
- Reward comes from the new labels only, no retraining: $r = \frac{1}{K}\sum_i y_i / \max(1, \mathrm{COUNT}_t(i))$, so rare classes score more.
- Posteriors are discounted ($\gamma = 0.9$) to follow rewards that change over time.

Results (10 imbalanced classification datasets, 10–20 rounds of 500–10,000 labels)

- Better than other meta-selectors on all datasets.
- On par with the best single strategy in accuracy on 9 of 10 (not Caltech256); better class diversity on 4 of 10; clearly better on CelebA.
- ALBL (accuracy-based reward) = random selection on all datasets.

Limits: classification only; large batches; non-stationarity left as future work.

For CARLA

- One round per flight; arms: every 3 s, random, least confident, most different, flickering plants, YOLO vs. T-Rex2 disagreement, under-covered altitude × light.

- *(my proposal)* Describe each labeled tile by binary attributes (contains cardilla, missed plant, false positive, altitude × light cell) and use TAILOR's multi-label reward unchanged.
- Measure: labels needed to reach a target mAP50, bandit vs. each arm vs. random.

2. Simple Supervision Is Hard to Beat — arXiv 2026

Problem

- Source-free adaptation of a detector with Mean Teacher pseudo-labels; a small random subset of target images is labeled. How to use those labels?

Method

- RTSM: keep self-training unchanged and add the ordinary detection loss on the labeled images.
- Compare against 10 plugins that use labels to steer pseudo-labels (selection, object completion, optimization control).

Results (Cityscapes → Foggy / BDD100K; DINO and DETA; 4 SFDA methods; 1–10% labels)

- RTSM: +1.7 to +18.3 AP50 over pure self-training, in every setting (e.g. DDT 53.7 → 58.7 with 5% labels).
- No plugin consistently beats RTSM; the best one changes with the method.
- Worst case: multi-view hard recovery −13.6 AP50 (PETS, BDD100K).
- More labels → plugins matter less (average effect +0.10 at 1%, −0.86 at 5%).
- Most stable: soft multi-view foreground revival (+0.2 to +0.8).

Limits: preprint; transformer detectors on street scenes; labels chosen at random, not actively.

For CARLA

- Training recipe each round: pseudo-labels + new labeled tiles as plain examples.
- Baseline the bandit must beat: random tiles + RTSM.
- Multi-frame filtering is close to "multi-view" plugins → *(my proposal)* use it as a soft signal, not as hard pseudo-labels.

3. On Warm-Starting — NeurIPS 2020

Problem

- Continuing training from the previous model is faster but generalizes worse, with the same training accuracy.

Results (50% → 100% of data; no augmentation, no LR schedule)

- ResNet-18 CIFAR-10: 56.2% → 51.7% (SGD), 78.0% → 74.4% (Adam).
- ResNet-18 CIFAR-100 (Adam): 41.4% → 35.0%. Logistic regression (convex): barely affected.
- Online (1,000 images per round): the gap grows as data arrive.
- Tuning LR, batch size or regularization closes the gap only by losing the speed benefit.

Fix: Shrink & Perturb

- Before each round: $\theta \leftarrow \lambda\theta + p$, with $0 < \lambda < 1$ and small noise p.
- $\lambda = 0.6$: same accuracy as training from scratch, much faster. Smaller λ can beat it.
- From pre-trained weights ($\lambda = 0.3$): at least as good as the better of warm start and scratch.

Limits: classification; no augmentation or schedules, which YOLO uses.

For CARLA

- Every flight = one retrain; from-scratch training every round is too slow on one GPU.

- CARLA's YOLO starts from COCO weights (pre-training case) and then gets flights (online case).
- *(my proposal)* Compare warm start, S&P (λ = 0.3, 0.6) and restart from COCO: mAP50 and GPU time per round.

4. LUF-net — Plant Phenomics 2026

Problem

- Drone colours shift with sun, clouds and the camera's automatic exposure, so flights are not comparable.

Method

- Light U-net (0.95M parameters) in Lab colour space.
- Metadata (aperture, exposure time, ISO + light-sensor irradiance) modulates features with FiLM: $F' = F(1 + \gamma(V)) + \beta(V)$.
- Ground truth from a 24-patch colour chart; 230 images, 58,292 patches, DJI Mavic 3M, 6–40 m, four light conditions.

Results (MAPE)

- Gray-world 14.6% · image-only U-net 9.31% · exposure only 4.95% · irradiance only 4.10% · full 3.89%.
- Sunny is hardest (5.30%), overcast easiest (3.53%).
- Error no longer depends on acquisition settings (r from 0.20–0.28 down to 0.07–0.08).

Limits: colour accuracy only, never detection; needs a colour chart; crops, still images, one drone.

For CARLA

- Exposure-only version needs no extra hardware if the DJI `.SRT` logs ISO, shutter and aperture (to check).
- The same telemetry gives each tile its altitude × light cell for the TAILOR reward.
- About 0.4 s per 1080p frame: fine offline.
- *(my proposal)* Train on sunny, test on overcast: none vs. photometric augmentation vs. LUF-net.

Putting it together

- One round: upload + telemetry ⟶ colour correction ⟶ detect on tiles ⟶ TAILOR picks k tiles ⟶ person labels ⟶ reward updates bandit ⟶ Shrink & Perturb ⟶ RTSM fine-tune ⟶ evaluate on held-out flights (worst condition).
- Shared lesson: simple signals win (label reward beats accuracy reward; plain supervision beats pseudo-label tricks; a rescaling beats hyperparameter tuning). Always report the simple baselines.

Questions for the supervisor

- Is a bandit enough RL, or is a learned policy (e.g. MGRAL, ICASSP 2026) expected?
- Training GPU and VRAM?
- How many flights and labeled tiles exist? Can 1–2 flights be fully labeled to simulate the loop?
- Which drone, are `.SRT` files uploaded, and can a colour chart be placed in a few flights?